Recent Advances in Missing Value Imputation: A Short Survey of State-of-the-Art Techniques

Hyejin Baek

bhj4208@gmail.com

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To Do List from Previous Seminar

23.10.23

- **IEIE SPC 투고 진행하지 않음**
  - ✓ Survey 논문에 대한 dynamic, datawig, zero, knn에 대한 앙상블 실험 진행
- **Survey 논문에 proposed method를 추가하여 다른 저널에 논문 투고 진행**

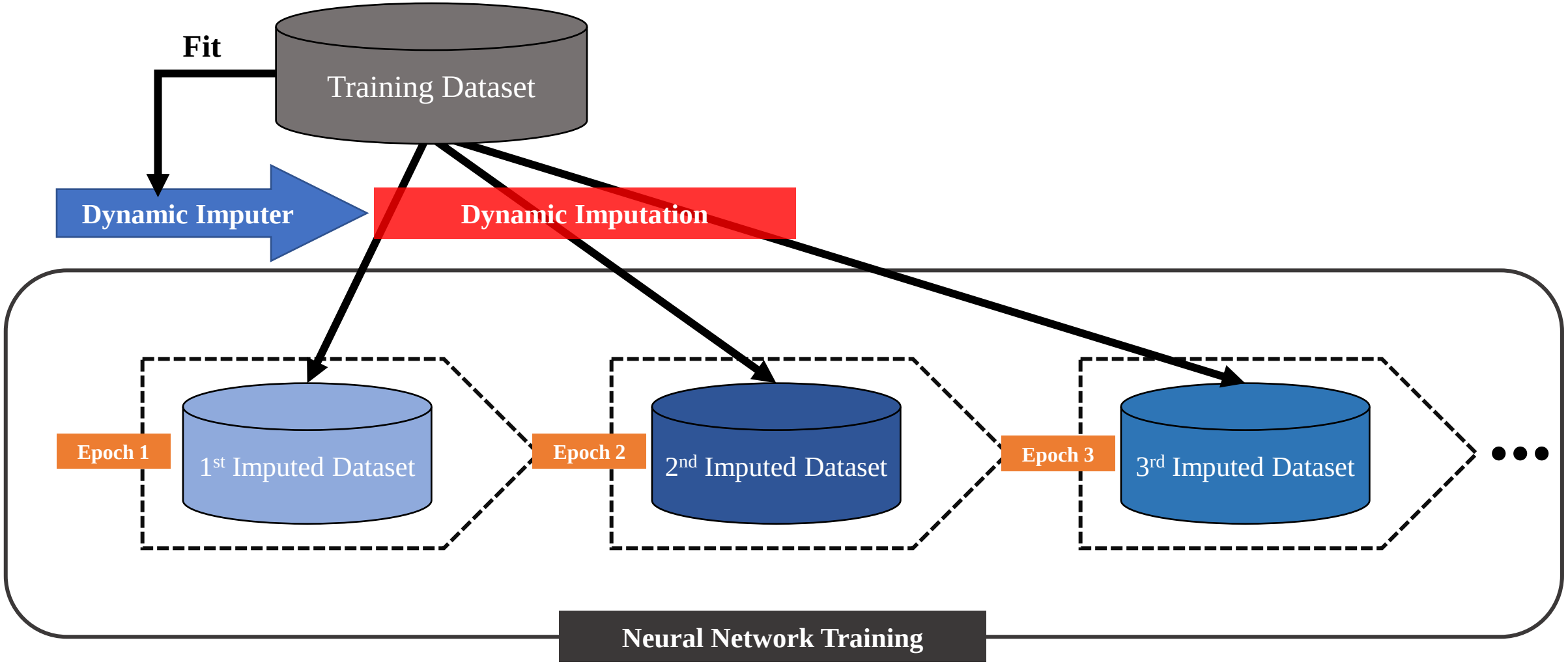
Texonomy

Imputation Method

Machine Learning-based Imputation	Deep Learning-based Imputation	Ensemble-based Imputation
K-means centroid [29]	DataWig [4]	rMisbeta [35]
Feature-wise selection [31]	MLP & DBN [20]	ZIPLN [1]
Takagi-Sugeno (TS) modeling [17]	End-to-end model [44]	Random forest-based + GRU [14]
GRLSR [5]	DAE [32]	Multiple singe imputation [46]
FAIA [10]	VISL [24]	
Clustering & top K nearest neighbor [8]	DeepIn [18]	
Query selection [11]	IFGAN [28]	
MLP [15]	Autoencoder [9, 7]	
Purity-based K nearest neighbors [6], [47], [48]	NAOMI [21]	
	Dynamic[12]	
	CMSDAE [34]	
	GRUF [40]	
	MGAIN [45]	

Dynamic imputation method[1]

✓ Unlike conventional approaches, the dynamic imputation method performs a new missing value imputation with a dynamic imputer for each training epoch.



DataWig[2]

- ✓ The imputation model in DataWig follows the approach of MICE, also referred to as fully conditional specification.
- ✓ For each to-be-imputed column(output column), the user can specify the columns which might contain useful information for imputation(input column)
- ✓ DataWig has a number of features that help to automate end-to-end imputation for practitioners.
- ✓ The data types are detected using heuristics and the corresponding features are learned automatically during the training of the imputation model.
- ✓ All hyperparameters and neural architectures are optimized using random search.

Experimental Settings

- ✓ Dataset : UCI Repository, total 10
- ✓ Dataset description

Table 1: Datasets

Dataset	Domain	Instances	Attributes	Type
breast [3]	Life	286	9	Categorical
heart [3]	Life	303	14	Categorical
adult [4]	Social	32561	15	Categorical
wine [5]	Social	178	13	Integer, Real
spambase [10]	Computer	4601	57	Integer, Real
chess [11]	Game	28056	6	Categorical
post [9]	Life	90	8	Categorical
liver [8]	Life	345	7	Categorical
abalone [7]	Life	4177	8	Categorical
lymph [7]	Life	148	18	Categorical

Experimental Settings

- ✓ Model architecture(2 hidden layer, 1 output layer)[1]

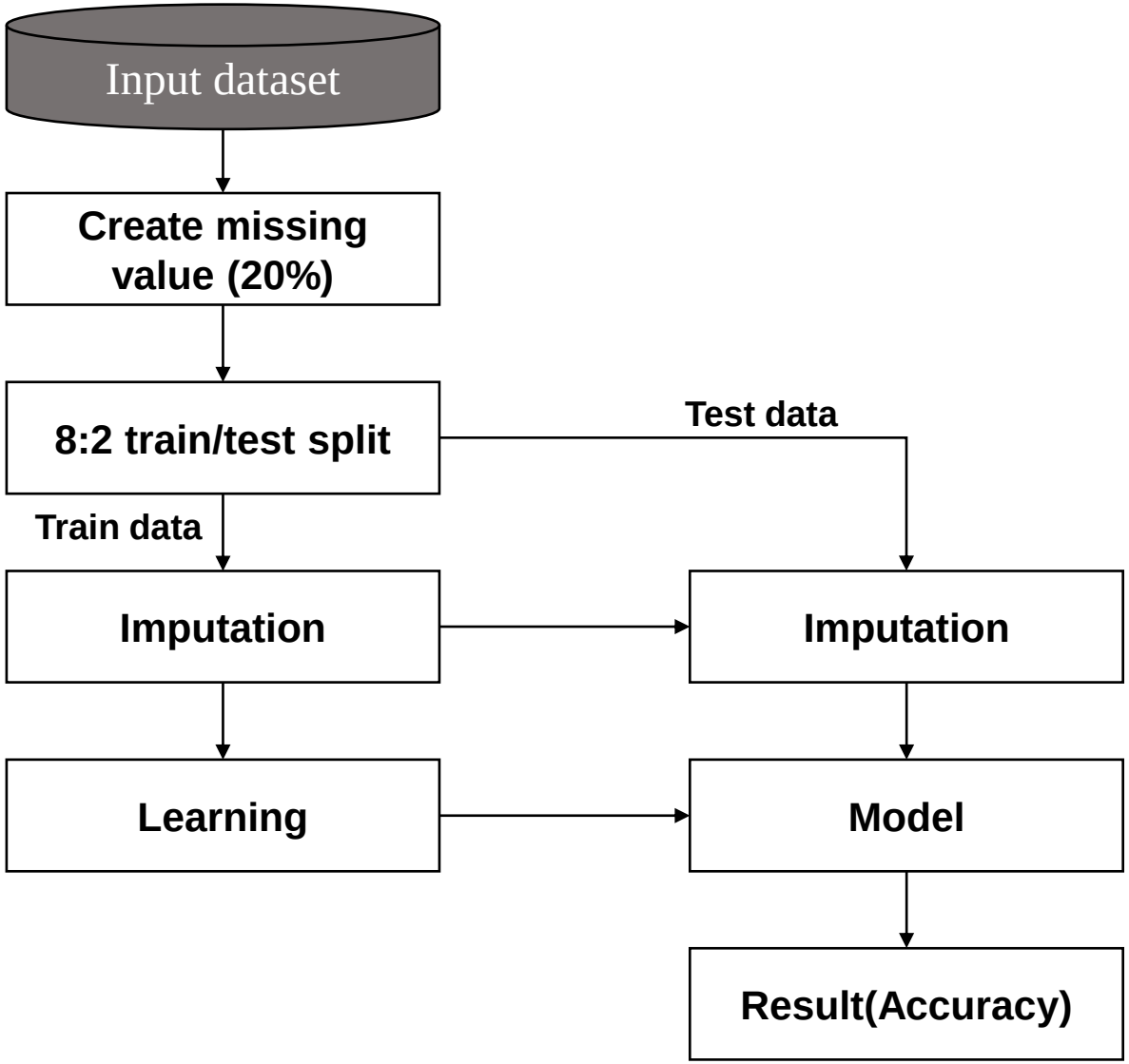
  → Optimizer : Adam

  → loss : BCE

  → Batch size / Epochs : 32/50

- ✓ Measure: Accuracy(8:2 train-test split, 10 iteration)

Experiment Process



Experimental Results

✓ Accuracy

Accuracy	breast	heart	adult	wine	spambase	chess	post patient	liver	abalone	lymphography
Datawig	0.7035 ± 0.0659	0.3787 ± 0.0982	0.7692 ± 0.0046	0.3833 ± 0.0928	0.9051 ± 0.0082	0.0503 ± 0.0030	0.0944 ± 0.0660	0.0638 ± 0.0312	0.3121 ± 0.0164	0.0100 ± 0.0213
Dynamic	0.7053 ± 0.0542	0.5475 ± 0.0727	0.0850 ± 0.0045	0.4361 ± 0.0938	0.6032 ± 0.0129	0.2102 ± 0.0062	0.6222 ± 0.0955	0.1159 ± 0.0415	0.4867 ± 0.0089	0.3966 ± 0.0585
Knn	0.7263 ± 0.0445	0.4180 ± 0.0820	0.7705 ± 0.0075	0.3833 ± 0.0928		0.0503 ± 0.0030	0.1333 ± 0.0619	0.0638 ± 0.0312	0.3096 ± 0.0110	0.4400 ± 0.0879
Zero	0.7053 ± 0.0651	0.3770 ± 0.0529	0.7727 ± 0.0050	0.3833 ± 0.0928		0.0503 ± 0.0030	0.1167 ± 0.1067	0.0638 ± 0.0312	0.3127 ± 0.0129	0.4267 ± 0.1181
zero+knn	0.7009 ± 0.0545	0.3959 ± 0.0880	0.7715 ± 0.0053	0.3833 ± 0.0928	0.9172 ± 0.0160	0.0503 ± 0.0030	0.1056 ± 0.0494	0.0638 ± 0.0312	0.3102 ± 0.0142	0.9822 ± 0.0307
zero+dynamic	0.6886 ± 0.0630	0.7852 ± 0.0630	0.7864 ± 0.0211	0.7958 ± 0.1693	0.7555 ± 0.1527	0.9503 ± 0.0032	0.9583 ± 0.0552	0.9362 ± 0.0312	0.6906 ± 0.0116	0.9900 ± 0.0213
zero+datawig	0.6930 ± 0.0525	0.4074 ± 0.0953	0.7706 ± 0.0043	0.3833 ± 0.0928		0.0503 ± 0.0030	0.1417 ± 0.0775	0.0638 ± 0.0312	0.3104 ± 0.0116	0.0100 ± 0.0213
knn+dynamic	0.7035 ± 0.0418	0.7967 ± 0.0512	0.7891 ± 0.0239	0.7958 ± 0.1695	0.7539 ± 0.1511	0.9502 ± 0.0029	0.9528 ± 0.0474	0.9355 ± 0.0312	0.6901 ± 0.0133	0.9900 ± 0.0213
knn+datawig	0.7298 ± 0.0405	0.4205 ± 0.1012	0.7699 ± 0.0050	0.3833 ± 0.0928		0.0503 ± 0.0030	0.1194 ± 0.0901	0.0638 ± 0.0312	0.3148 ± 0.0144	0.0100 ± 0.0213
dynamic+datawig	0.6991 ± 0.0480	0.7943 ± 0.0625	0.7883 ± 0.0228	0.8028 ± 0.1752		0.9505 ± 0.0031	0.9556 ± 0.0544	0.9355 ± 0.0312	0.6911 ± 0.0099	0.9900 ± 0.0213
dynamic+datawig+zero	0.7023 ± 0.0483	0.7787 ± 0.0567	0.7966 ± 0.0220	0.8500 ± 0.1592		0.9510 ± 0.0032	0.9722 ± 0.0448	0.9343 ± 0.0319	0.6868 ± 0.0127	0.9878 ± 0.0219
dynamic+datawig+knn	0.6931 ± 0.0409	0.7721 ± 0.0633	0.7962 ± 0.0227	0.8537 ± 0.1615		0.9505 ± 0.0030	0.9667 ± 0.0444	0.9348 ± 0.0313	0.6879 ± 0.0157	0.9867 ± 0.0237
datawig+zero+knn	0.7070 ± 0.0395	0.4191 ± 0.0832	0.7710 ± 0.0051	0.3833 ± 0.0928		0.0503 ± 0.0030	0.1111 ± 0.0896	0.0638 ± 0.0312	0.3120 ± 0.0143	0.0100 ± 0.0213
dynamic+zero+knn	0.6877 ± 0.0654	0.7743 ± 0.0610	0.7954 ± 0.0217	0.8481 ± 0.1581		0.9502 ± 0.0031	0.9685 ± 0.0398	0.9343 ± 0.0321	0.6860 ± 0.0145	0.9889 ± 0.0217
all	0.7034 ± 0.0410	0.7725 ± 0.0584	0.7991 ± 0.0200	0.8757 ± 0.1463		0.9502 ± 0.0034	0.9667 ± 0.0427	0.9344 ± 0.0304	0.6821 ± 0.0164	0.9892 ± 0.0216

✓ 파란색 글씨 : Multi-class dataset
✓ Spambase 데이터셋 datawig imputation method 이슈로 인해 실험 진행 후 추가 예정
✓ Accuracy에서는 데이터셋의 형태가 다양하여 특정 method가 뛰어나다고 판단하기 힘듬

Experimental Results

✓ RMSE

RMSE	breast	heart	adult	wine	spambase	chess	post patient	liver	abalone	lymphography
Datawig	0.5082 ± 0.0305	1.3107 ± 0.1314	0.4318 ± 0.0029	1.1700 ± 0.1205	0.6524 ± 0.0103	8.3572 ± 0.0578	1.3656 ± 0.1130	4.0988 ± 0.4643	0.8707 ± 0.0173	1.5728 ± 0.0794
Dynamic	0.4478 ± 0.0279	0.3501 ± 0.0254	0.0670 ± 0.0020	0.1821 ± 0.0243	0.3009 ± 0.0094	0.2085 ± 0.0004	0.4179 ± 0.0163	0.2276 ± 0.0048	0.2475 ± 0.0099	0.3139 ± 0.0094
Knn	0.4481 ± 0.0360	1.1591 ± 0.0955	0.4209 ± 0.0048	1.1700 ± 0.1205	0.2325 ± 0.0113	8.3572 ± 0.0578	1.4384 ± 0.1484	4.0988 ± 0.4643	0.7763 ± 0.0115	0.7558 ± 0.0629
Zero	0.4458 ± 0.0425	1.2362 ± 0.1267	0.4141 ± 0.0030	1.1700 ± 0.1205	0.2687 ± 0.0086	8.3572 ± 0.0578	1.4179 ± 0.1492	0.0638 ± 0.0312	0.7870 ± 0.0099	0.7749 ± 0.1105
zero+knn	0.4473 ± 0.0309	1.1768 ± 0.1198	0.4124 ± 0.0031	1.1700 ± 0.1205	0.2352 ± 0.0094	8.3572 ± 0.0578	1.3879 ± 0.1642	4.0988 ± 0.4643	0.7789 ± 0.0096	0.2340 ± 0.0234
zero+dynamic	0.0919 ± 0.0129	0.1136 ± 0.0132	0.1133 ± 0.0023	0.0786 ± 0.0278	0.1261 ± 0.0031	0.0621 ± 0.0032	0.2138 ± 0.0199	0.0316 ± 0.0067	0.0550 ± 0.0042	0.1776 ± 0.0135
zero+datawig	0.4561 ± 0.0185	1.1711 ± 0.1356	0.4130 ± 0.0026	1.1700 ± 0.1205		8.3572 ± 0.0578	1.4223 ± 0.1510	4.0997 ± 0.4648	0.7825 ± 0.0093	1.5728 ± 0.0794
knn+dynamic	0.0929 ± 0.0106	0.0946 ± 0.0107	0.1178 ± 0.0032	0.0849 ± 0.0246	0.1305 ± 0.0051	0.0596 ± 0.0036	0.2190 ± 0.0129	0.0313 ± 0.0065	0.0482 ± 0.0050	0.1752 ± 0.0143
knn+datawig	0.4433 ± 0.0195	1.1621 ± 0.1175	0.4204 ± 0.0025	1.1700 ± 0.1205		8.3572 ± 0.0578	1.4449 ± 0.1590	4.0988 ± 0.4643	0.7792 ± 0.0078	1.5728 ± 0.0794
dynamic+datawig	0.1004 ± 0.0162	0.1046 ± 0.0125	0.1125 ± 0.0038	0.0908 ± 0.0255		0.0609 ± 0.0027	0.2183 ± 0.0117	0.0325 ± 0.0079	0.0549 ± 0.0045	0.1789 ± 0.0108
dynamic+datawig+zero	0.1276 ± 0.0111	0.1289 ± 0.0176	0.1381 ± 0.0030	0.1050 ± 0.0340		0.0768 ± 0.0022	0.2862 ± 0.0187	0.0444 ± 0.0097	0.0680 ± 0.0055	0.2341 ± 0.0210
dynamic+datawig+knn	0.1166 ± 0.0173	0.1153 ± 0.0132	0.1329 ± 0.0029	0.1070 ± 0.0305		0.0731 ± 0.0033	0.2864 ± 0.0187	0.0409 ± 0.0078	0.0563 ± 0.0039	0.2370 ± 0.0210
datawig+zero+knn	0.4475 ± 0.0203	1.1560 ± 0.1164	0.4136 ± 0.0029	1.1700 ± 0.1205		8.3572 ± 0.0578	1.4147 ± 0.1435	4.0988 ± 0.4643	0.7779 ± 0.0097	1.5728 ± 0.0794
dynamic+zero+knn	0.1061 ± 0.0129	0.1213 ± 0.0141	0.1336 ± 0.0042	0.0979 ± 0.0314		0.0738 ± 0.0023	0.2839 ± 0.0155	0.0432 ± 0.0088	0.0552 ± 0.0051	0.2354 ± 0.0201
all	0.1280 ± 0.0139	0.1293 ± 0.0155	0.1444 ± 0.0038	0.1107 ± 0.0340		0.0801 ± 0.0035	0.3270 ± 0.0202	0.0454 ± 0.0102	0.0620 ± 0.0054	0.2703 ± 0.0274

✓ 파란색 글씨 : Multi-class dataset
✓ Spambase 데이터셋 datawig imputation method 이슈로 인해 실험 진행 후 추가 예정
✓ RMSE에서는 Zero+Dynamic의 성능이 뛰어난 것으로 확인함.

Experimental Results

- ✓ Dynamic method가 포함된 앙상블 실험에서 multi-class인 데이터셋(heart, wine, chess, post-patient, liver, abalone, lymphography)에서 dynamic 모델 안에서의 클래스 불일치 문제 발생하여 binary class로 변경 후 진행하여 성능이 지나치게 향상함.

  → Multi-class는 고려하지 못한 부분이었기에 현재 분석 중이며 교수님과 논의 필요

  → 본 논문의 실험 목적은 데이터셋의 특징별 어떤 imputation method가 효과적인지 파악을 목표로 하고자 함.

- ✓ Spambase dataset에서 Datawig method가 포함된 앙상블 실험에서 이슈 발생함.

  → 완료 후 추가예정

To Do List

- ✓ 실험 결과에 대한 교수님과 논의 필요
- ✓ Survey 추가 실험 결과 논문 작성

Reference

[1] Han, Jongmin, and Seokho Kang, “Dynamic imputation for improved training of neural network with missing values” Expert Systems with Applications, 2022

[2] Biessmann, and Felix, “DataWig:Missing Value Imputation for tables” Journal of Machine Learning, 2019

Thank you